

EmissionGate

Carbon at merge time. An agent that puts the kgCO2e of cloud and AI infrastructure into the pull request — and leaves the decision to a human.

EmissionGate — carbon check

 Acknowledgement required

+683.5

kgCO2e per year • range 263–1,104 (assumed load)

gpu_fleet: +2 × g5.2xlarge

Suggestion: weekday schedule (plan ✓)

Carbon is committed at merge and seen months later



Too late

Cloud carbon reports arrive monthly, by service and region — never per resource, never per change.



Wrong ranking

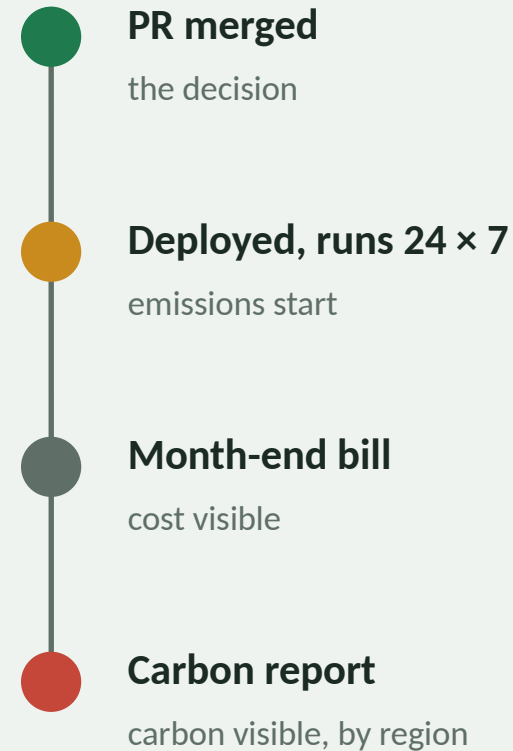
Cost tools optimise a different list: archiving 40 TB of logs saves money and ≈ 0 kgCO₂e; Graviton is the reverse.



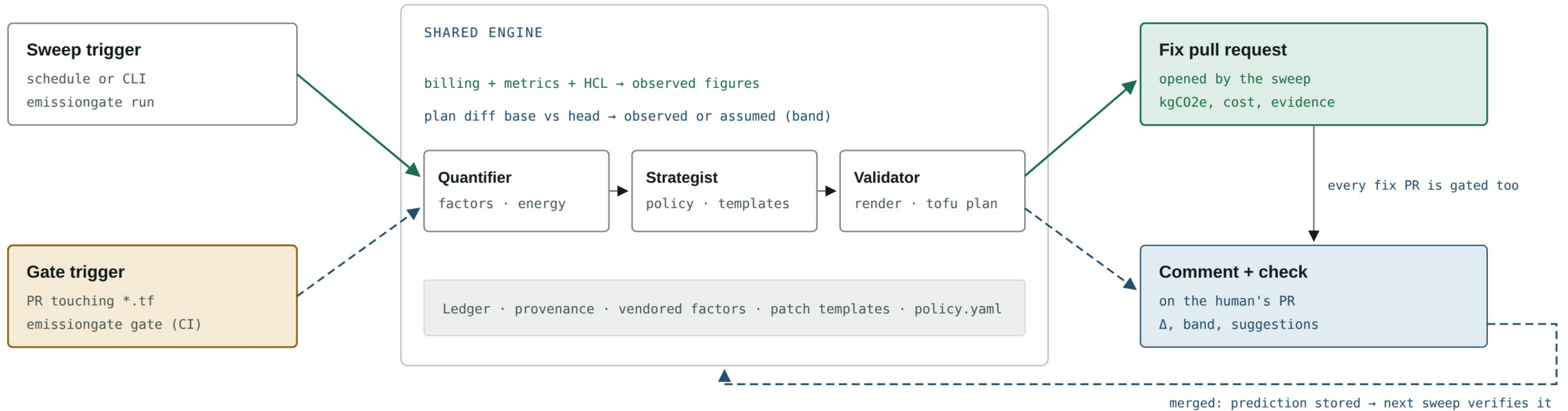
Waste runs for months

An idle staging GPU fleet emits 401 kgCO₂e a year before anyone links it to the PR that created it.

Where the decision is made vs where carbon appears



Two modes, one engine



Sweep

Finds waste in running infrastructure and drafts the fix as a PR, with kgCO2e, cost and evidence.

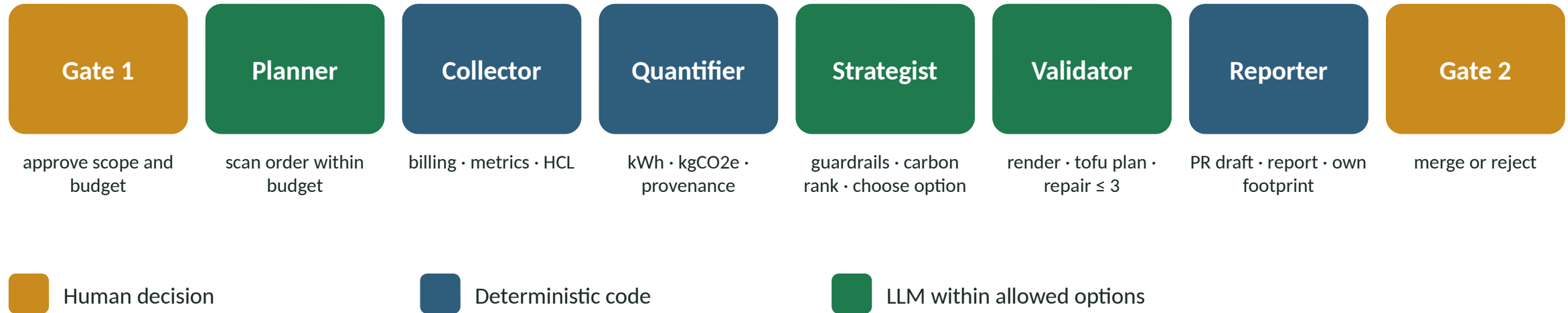
Gate

Checks every infrastructure PR before merge; posts the delta and validated lower-carbon suggestions.

One set of numbers

Both modes share every component that produces a figure, so a comment and a report always agree.

How it works: deterministic numbers, bounded judgement



Deterministic core — every published number

Energy model, vendored factors, policy, ranking, patch templates, ledger, SCI.

It cannot import the LLM package — a test fails the build if it does.

LLM perimeter — choices and prose only

Local gpt-oss-20b picks among options the code allows, repairs a failed plan, writes the narrative.

Schema-validated JSON with no numeric fields; one re-ask, then the rule-based default.

Humans decide: the agent informs, proposes and waits



Cannot apply, destroy or touch AWS

no apply path in the code; plans run with dummy credentials



Cannot start itself

Gate 1: a human approves scope and budget; recorded in the ledger



Cannot merge

every change is a pull request; Gate 2 is a human review



Refuses protected resources

DR, month-end and regulatory traps are blocked before any decision

Above 100 kgCO₂e/yr the PR check fails until a human acts



Commit a suggestion

each suggestion is rendered from a template and proven with tofu plan



Or accept the increase

label eg/carbon-accepted plus a written reason; who and why go to the ledger



Below 25 kgCO₂e/yr

pass, no noise; 25–100 is a warning

Methodology: published coefficients, pinned and tested

Per resource, Cloud Carbon Footprint model

```
cpu_w = vCPU × (min + u × (max - min))  
gpu_w = GPUs × (min + u × (max - min))  
mem_w = extra GiB × memory coefficient  
kWh    = (cpu + gpu + mem) × PUE 1.135 × h  
kgCO2e = kWh × regional g/kWh ÷ 1000
```

Location-based, GHG Protocol Scope 2. Annual regional factors for totals; half-hourly UK grid data only where it changes a decision (time-shifting).

Sources

CCF coefficients vendored at commit f584c549 (Apache-2.0) · UK Carbon Intensity (CC BY 4.0) · synthetic estate, seed 42

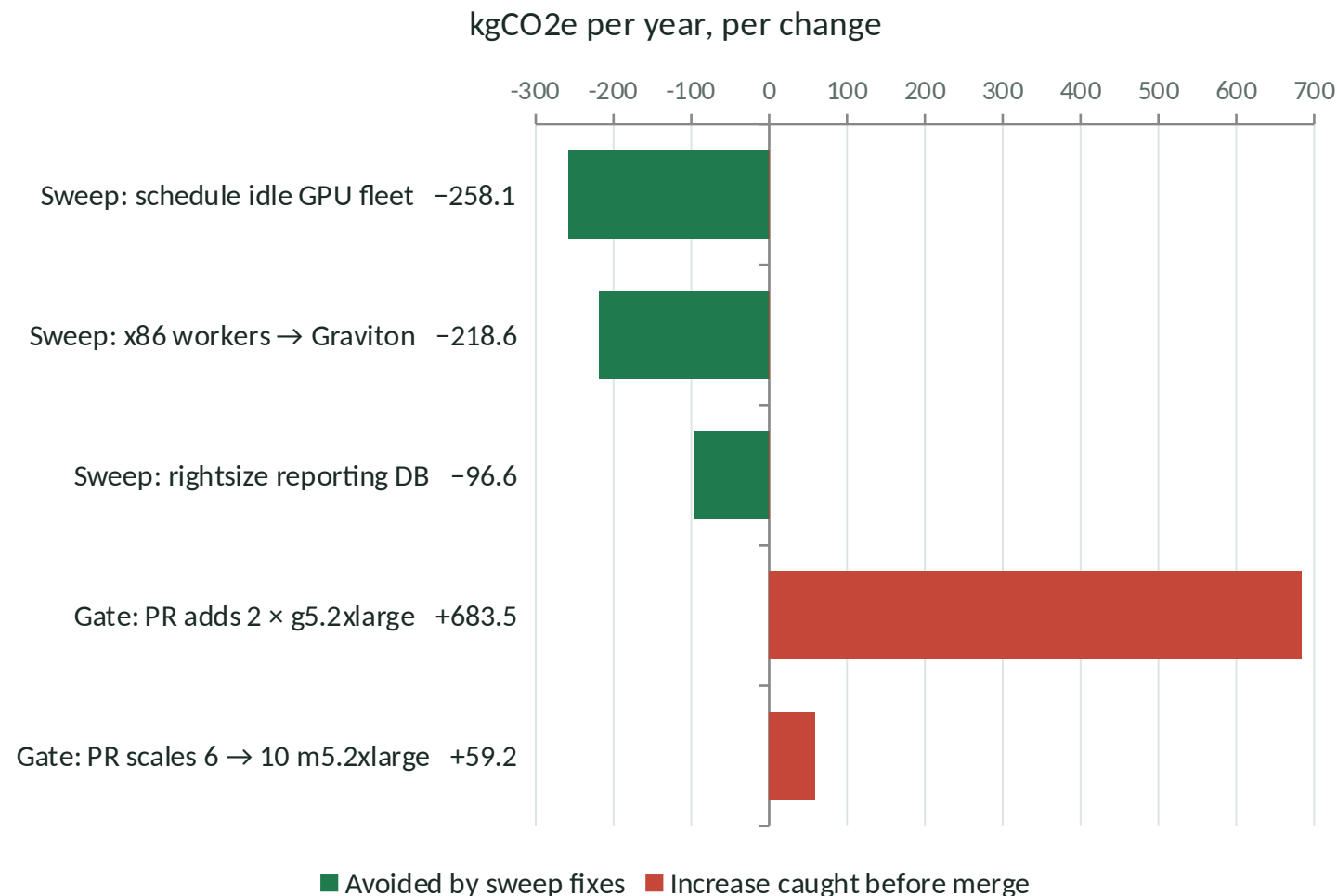
New resources are assumed

no history → CCF's 50% utilisation default, published as a 10–90% range and labelled 'assumed'

Verified, not asserted

golden values G1–G7 reproduced by tests to 3 decimals; provenance on every figure; append-only ledger

Results on the synthetic estate



573

kgCO2e/yr avoided by three sweep fixes

+684

kgCO2e/yr stopped at review (range 263–1,104)

≈ 0

kgCO2e from archiving 40 TB of logs — a cost saving, not a carbon one

Worked examples on the estate's own resources, reproduced by the test suite. Run metrics (precision, recall, trap violations, the agent's own energy) are in the README.

Impact: carbon becomes a review comment, not a report

	Today	With EmissionGate
When	monthly, after the fact	at merge, before the resource exists
Granularity	service and region	per resource, per change
Who sees it	a sustainability team	the engineer who can change it
What happens	a dashboard	a validated fix, or a written acknowledgement
Evidence	totals	provenance on every figure, append-only ledger



Prevention compounds

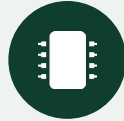
Per-change figures are tens to hundreds of kgCO₂e a year — that is what the coefficients say. The gate sees every infrastructure PR, so each one caught is avoided for the resource's whole life, with no clean-up project later.

Efficiency choices: a small agent that measures itself



Deterministic maths

numbers come from code, so the model is called rarely and briefly



One local open model

gpt-oss-20b (Apache-2.0), ~3.6B active parameters, low reasoning by default



No LLM in CI

the gate is arithmetic over a plan and finishes in seconds



Templates, not generated code

fewer failed plans, fewer tokens, reviewable diffs



Hard budgets + offline mode

caps on calls, tokens, repairs, time; an LLM-free baseline to compare against



Nothing deployed

one process that runs and exits; providers from a local mirror

Own footprint, every run: CodeCarbon wraps the whole run, model included → kWh, kgCO₂e per run (SCI, ISO/IEC 21031), tokens, and payback ratio = kgCO₂e/yr proposed ÷ kgCO₂e emitted. Labelled 'estimated' where the laptop has no power counters.

Roadmap

● This submission

- Sweep, offline and with a local LLM
- PR gate in GitHub Actions with acknowledgement
- Guardrails, five fix templates, plan validation
- Own footprint in every run report

● Next

- Measured model bake-off (gpt-oss vs Qwen)
- Real fix PRs; rejection labels become rules
- Live grid API for time-shifting
- Calibrate assumed loads against what deploys

● Later

- Azure and Google Cloud coefficients
- Same state machine on Step Functions
- Organisation-wide thresholds and reporting

Code, design document, augmentation log and demo: see the repository README.